



Nutrition

Functions of Different Nutrients

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Learning Outcomes:

- Identify the role of carbohydrates in the body.
- Recognize the functions of different types of lipids.
- Categorize the proteins according to their functions.
- Summarize the role of different minerals in the body.
- Describe the physiological role of water soluble and fat soluble vitamins.

Carbohydrates:

- Carbohydrates function mainly as a source of chemical energy for generating ATP needed to drive metabolic reactions.
- Only a few carbohydrates are used for building structural units. One example is **deoxyribose**, a type of sugar that is a building block of deoxyribonucleic acid (DNA).
- Glucose is the body's preferred source for synthesizing ATP.



The Fate of Glucose:

Because glucose is the body's preferred source for synthesizing ATP, its use depends on the needs of body cells, which include the following:

1. **ATP production:** In body cells that require immediate energy, glucose is oxidized to produce ATP.
2. **Amino acid synthesis:** Cells throughout the body can use glucose to form several amino acids, which then can be incorporated into proteins.

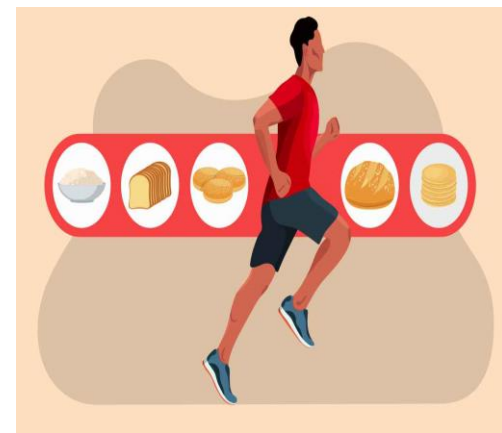
The Fate of Glucose:

3. Glycogen synthesis: Hepatocytes and muscle fibers can perform glycogenesis, in which hundreds of glucose monomers are combined to form glycogen.

4. Triglyceride synthesis: When the glycogen storage areas are filled up, hepatocytes can transform the glucose to triglycerides. Triglycerides then are deposited in adipose tissue, which has virtually unlimited storage capacity!

CLINICAL CONNECTION / Carbohydrate Loading:

- The amount of glycogen stored in the liver and skeletal muscles varies and can be completely exhausted during long-term athletic endeavors.
- Thus, many marathon runners and other endurance athletes follow a precise exercise and dietary regimen that includes eating large amounts of complex carbohydrates, such as pasta and potatoes, in the 3 days before an event.
- This practice, called **carbohydrate loading**.
- For athletic events lasting more than an hour, carbohydrate loading has been shown to increase an athlete's endurance. This is due to increased glycogenolysis, which results in more glucose that can be catabolized for energy.



Lipids

- Most lipids are insoluble in polar solvents such as water; they are hydrophobic.
- To become more soluble in blood plasma, other lipid molecules join with hydrophilic protein molecules. The resulting lipid–protein complexes are termed **lipoproteins**.
- Each lipoproteins has different functions, but all are essentially transport vehicles. They make lipids available when cells need them or remove lipids from circulation when they are not needed.

- Lipoproteins are categorized and named mainly according to their density, there are four major classes:

1) **Chylomicrons**: which form in mucosal epithelial cells of the small intestine, transport dietary (ingested) lipids to adipose tissue for storage.

2) **VLDLs** (very-low-density lipoproteins): transport triglycerides synthesized in hepatocytes to adipocytes for storage.

3) **LDLs** (low-density lipoproteins): carry about 75% of the total cholesterol in blood and deliver it to cells throughout the body for use in repair of cell membranes and synthesis of steroid hormones and bile salts.

- When present in excessive numbers, **LDLs** also deposit cholesterol in and around smooth muscle fibers in arteries, forming fatty plaques that increase the risk of coronary artery disease. For this reason, the cholesterol in LDLs, called LDL-cholesterol, is known as “**bad**” **cholesterol**.



4) HDLs (high-density lipoproteins): remove excess cholesterol from body cells and the blood and transport it to the liver for elimination.

- Because **HDLs** prevent accumulation of cholesterol in the blood, a high HDL level is associated with decreased risk of coronary artery disease. For this reason, HDL-cholesterol is known as “**good**” **cholesterol**.

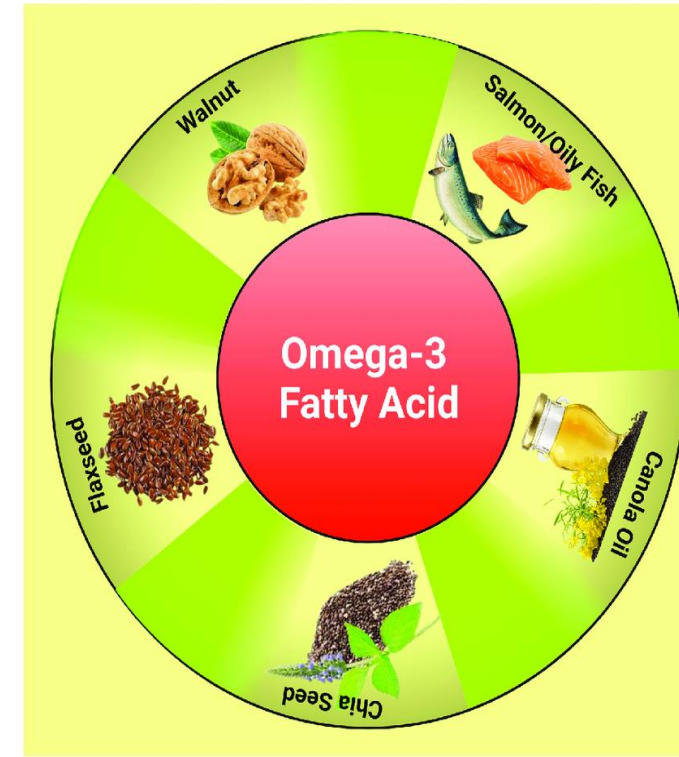
The Fate of Lipids

- Lipids, like carbohydrates, may be oxidized to produce ATP. If the body has no immediate need to use lipids in this way, they are stored in adipose tissue (fat depots) throughout the body and in the liver.
- A few lipids are used as structural molecules or to synthesize other essential substances, Some examples include
 - ❑ **Phospholipids**, which are constituents of plasma membranes.
 - ❑ **Lipoproteins**, which are used to transport cholesterol throughout the body.
 - ❑ **Thromboplastin**, which is needed for blood clotting.
 - ❑ **Myelin sheaths**, which speed up nerve impulse conduction.

Types of Lipids in the Body	
TYPE OF LIPID	FUNCTIONS
Fatty acids	Used to synthesize triglycerides and phospholipids or catabolized to generate adenosine triphosphate (ATP).
Triglycerides (fats and oils)	Protection, insulation, energy storage.
Phospholipids	Major lipid component of cell membranes.
Steroids <i>Cholesterol</i> <i>Bile salts</i> <i>Vitamin D</i> <i>Adrenocortical hormones</i> <i>Sex hormones</i>	Minor component of all animal cell membranes; precursor of bile salts, vitamin D, and steroid hormones. Needed for digestion and absorption of dietary lipids. Helps regulate calcium level in body; needed for bone growth and repair. Help regulate metabolism, resistance to stress, and salt and water balance. Stimulate reproductive functions and sexual characteristics.
Eicosanoids (prostaglandins and leukotrienes)	Have diverse effects on modifying responses to hormones, blood clotting, inflammation, immunity, stomach acid secretion, airway diameter, lipid breakdown, and smooth muscle contraction.
Other lipids <i>Carotenes</i> <i>Vitamin E</i> <i>Vitamin K</i> <i>Lipoproteins</i>	Needed for synthesis of vitamin A (used to make visual pigments in eye); function as antioxidants. Promotes wound healing, prevents tissue scarring, contributes to normal structure and function of nervous system, and functions as antioxidant. Required for synthesis of blood-clotting proteins. Transport lipids in blood, carry triglycerides and cholesterol to tissues, and remove excess cholesterol from blood.

CLINICAL CONNECTION / Fatty Acids in Health and Disease :

- a group of fatty acids called essential fatty acids (EFAs) is essential to human health, they cannot be made by the human body and must be obtained from foods or supplements.
- Among the more important EFAs are omega-3 fatty acids, omega-6 fatty acids, and cis-fatty acids.
- Omega-3 and omega-6 fatty acids are polyunsaturated fatty acids that are believed to work together to promote health.
- They may have a protective effect against heart disease and stroke by lowering total cholesterol, raising HDL and lowering LDL.



CLINICAL CONNECTION / Fatty Acids in Health and Disease :

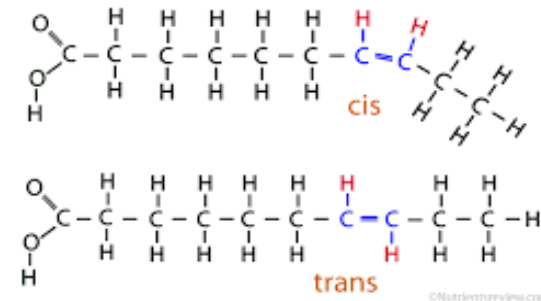
- Omega-3 and omega-6 fatty acids decrease bone loss reduce symptoms of arthritis, promote wound healing; improve certain skin disorders like eczema; and improve mental functions.
- Primary sources of omega-3 fatty acids include: fatty fish and walnuts.
- Primary sources of omega-6 fatty acids include most processed foods (cereals, breads, white rice), eggs, baked goods and meats (especially organ meats, such as liver).

CLINICAL CONNECTION / Fatty Acids in Health and Disease :

- Cis-fatty acids are nutritionally beneficial, but when they are heated, pressurized, and combined with a catalyst in a process called hydrogenation, they are changed to unhealthy trans-fatty acids.
- Hydrogenation is used by manufacturers to make vegetable oils solid at room temperature and less likely to turn rancid.
- If oil used for frying is reused, cis-fatty acids are converted to trans-fatty acids.
- Among the adverse effects of trans-fatty acids are an increase in total cholesterol, a decrease in HDL, an increase in LDL, and an increase in triglycerides. These effects, which can increase the risk of heart disease and other cardiovascular diseases.

% Daily Value*	
Total Fat 13g	20%
Saturated Fat 5g	25%
Trans Fat 2g	
Cholesterol 30mg	10%
Sodium 660mg	28%
Total Carbohydrate 31g	10%
Dietary Fiber 0g	0%
Sugars 5g	
Protein 5g	
Vitamin A 4%	* Vitamin C 2%
Calcium 15%	* Iron 4%
* Percent Daily Values are based on a 2,000 caloric diet. Your Daily Values may be higher or lower depending on your caloric needs.	

Cis- and Trans-Fatty Acids



Proteins

The Fate of Proteins:

- The active transport of amino acids into body cells is stimulated by insulinlike growth factors (IGFs) and insulin.
- Almost immediately after digestion, amino acids are reassembled into proteins.

Functions Of Proteins:

Different Types of Proteins have different functions:

Structural

Structural framework of various parts of body.

Examples: collagen in bone and other connective tissues; keratin in skin, hair.

Regulatory

Function as hormones that regulate various physiological processes; control growth and development; or as neurotransmitters

Examples: the hormone insulin (regulates blood glucose level).

Contractile

Allow shortening of muscle cells, which produces movement.

Examples: myosin; actin.

Functions Of Proteins:

Immunological



Aid responses that protect body against foreign substances and invading pathogens.

Examples: antibodies; interleukins.

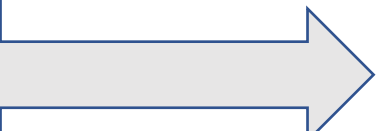
Transport



Carry vital substances throughout body.

Example: hemoglobin (transports most O₂ and some CO₂ in blood).

Catalytic



Act as enzymes that regulate biochemical reactions.

Examples: salivary amylase, ATPase.

Minerals vital to the body

Minerals	Importance
Calcium	Formation of bones and teeth, blood clotting , cellular motility and during cell division, normal muscle and nerve activity.
Phosphorus	Formation of bones and teeth , constitute a major buffer system of blood, muscle contraction and nerve activity, energy transfer (ATP), component of DNA and RNA.
Potassium	Generation and conduction of action potentials in neurons and muscle fibers.
Sulfur	Component of hormones and vitamins, needed for ATP production.
Sodium	Osmosis , buffer system, nerve and muscle action potential conduction .
Chloride	Acid–base balance of blood, HCl in stomach .
Magnesium	Normal functioning of muscle and nervous tissue, bone formation, Constituent of many coenzymes.

Minerals vital to the body

Minerals	Importance
Iron	Component of hemoglobin, reversibly binds O ₂
Iodide	Synthesize thyroid hormones
Manganese	Hemoglobin synthesis, growth, reproduction, lactation, bone formation
Copper	Synthesis of hemoglobin, components of coenzymes and enzymes.
Cobalt	Part of vitamin B ₁₂ , required for erythropoiesis
Zinc	Carbon dioxide metabolism, growth and wound healing, taste sensations and appetite, and normal sperm counts.
Fluoride	Improve tooth structure and inhibit tooth decay
Selenium	Synthesis of thyroid hormones, immune system, functions as antioxidant.
Chromium	Carbohydrate and lipid metabolism.

Functions of Fat Soluble Vitamins:

Vitamin	Functions
A	formation of light-sensitive pigments in photoreceptors of retina. Aids in growth of bones and teeth, Beta-carotene acts as antioxidant to inactivate free radicals
D	absorption of calcium and phosphorus from GI tract. Works with parathyroid hormone (PTH) to maintain Ca homeostasis.
E (tocopherols)	help form cell structures, especially membranes. formation of DNA, RNA, and red blood cells, wound healing and prevent scarring, contribute to normal structure and functioning of nervous system, Acts as antioxidant
K	Coenzyme essential for synthesis of several clotting factors by liver, including prothrombin.

Functions of Water Soluble Vitamins:

Vitamin	Functions
B1 (thiamine), B2 (riboflavin), Niacin (nicotinamide), B6 (pyridoxine), B12 (cyanocobalamin), Pantothenic acid and Biotin .	Act as coenzyme with other enzymes that are essential for metabolism and other reactions.
Folic acid (folate, folacin)	Component of enzyme systems synthesizing nitrogenous bases of DNA and RNA. Essential for normal production of red and white blood cells . Higher risk of neural tube defects in babies born to folate-deficient mothers.
C (ascorbic acid)	Promotes protein synthesis, including laying down of collagen in formation of connective tissue. As coenzyme, may combine with poisons, rendering them harmless until excreted. Works with antibodies , promotes wound healing, and functions as an antioxidant .

Questions and Comments?

